

Solve Problems with Unit Rates

Lesson 4-7

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

\$2.50 per pound means \$2.50 for every 1 pound

Unit Rate

A rate for just 1 of something, like cost for 1 item.

\$0.10 each vs \$0.15 each — the \$0.10 option is the better buy

Better Buy

The choice that costs less for each item.

\$3 per game < \$4 per game

Comparison

Looking at numbers to see which is bigger or smaller.

\$0.50 per token means each token costs \$0.50

Per Unit

For each single item.

120 miles in 3 hours is a rate (miles and hours are different units)

Rate

A ratio comparing two amounts with different units, like miles per hour.

 $5/3 = x/12 \rightarrow x = 20$

Proportion

A math sentence saying two ratios are equal. It helps find a missing number.

Key Ideas & Notes

- You're the new supply manager for the arcade!
- You need to order tokens, prizes, and snacks from different suppliers.
- Each supplier offers a different package deal.
- Your job is to find the unit rate for each deal and figure out which supplier gives you the best value for every item you need to order.
- Calculate the unit rate for each supplier's deal. Then determine which is the better buy for each item.

Think About It

- How can you compare deals when the quantities are different?
- What operation helps you find the cost for just one item?
- Does the biggest package always give the best unit rate?

My Notes

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

As supply manager you compare token deals like 500 for \$60 and 350 for \$38.50. You can't compare totals directly, so what does finding the cost 'per unit' let you do?

Level 1 support

Start here: Finding the cost per one token lets me compare deals fairly even when the package sizes differ.

I can't compare the totals because ____.

No puedo comparar los totales porque ____.

Finding the cost ____ lets me compare fairly.

Hallar el costo ____ me deja comparar de forma justa.

Word bank: unit rate per unit comparison rate

Listen for: Students explain the packages have different quantities, so the per-unit cost (unit rate) is needed to compare deals fairly.

Level 2

Does the biggest package always have the lowest unit rate? Predict, then explain how you could be tricked by package size alone.

- The biggest package is not always cheapest per unit because ____.
- Package size can trick you when ____.

EXPLORE

How did the unit rate help you decide the better buy between the two token suppliers in your table?

Level 1 support

Start here: I found the cost per token for each supplier and chose the lower one.

The unit rate for ____ is \$____ per token.

La tasa unitaria de ____ es \$____ por ficha.

The better buy is ____ because its unit rate is ____.

La mejor compra es ____ porque su tasa unitaria es ____.

Word bank: (unit rate) (better buy) (per unit) (comparison) (rate)

Listen for: Students divide price by quantity for each deal (e.g., 350 for \$38.50 = \$0.11 each) and choose the supplier with the lower unit rate as the better buy.

Level 2

One supplier had the lower price per token but a higher total. Explain why a lower unit rate, not a lower total, is what makes a deal the better buy.

- A lower total does not mean a better buy because ____.
- The unit rate is what matters because ____.

CONNECT

Printer A makes 150 tickets in 5 minutes and Printer B makes 200 in 8 minutes. The manager needs 600 tickets fast. How do unit rates decide the faster printer and the time needed?

Level 1 support

Start here: I find tickets per minute for each printer, pick the higher rate, then divide 600 by it.

Printer A prints ____ tickets per minute and Printer B prints ____.

La impresora A imprime ____ boletos por minuto y la B imprime ____.

The manager should use Printer ____; 600 tickets take ____ minutes.

El gerente debe usar la impresora ____; 600 boletos tardan ____ minutos.

Word bank: unit rate rate per unit comparison proportion

Listen for: Students find 30 and 25 tickets/min, choose Printer A (higher rate), and divide $600 / 30 = 20$ minutes.

Level 2

Here the higher unit rate is better, but for the token deals the LOWER unit rate was better. Explain why 'better' depends on what the rate measures.

- A higher rate is better when ____, but a lower rate is better when ____.
- It depends on whether the rate measures ____ or ____.

Guided Examples

Example 1

Which is the better buy: 6 bouncy balls for \$4.50 or 10 bouncy balls for \$8.00?

Solution: $\$4.50 \div 6 = \0.75 each. $\$8.00 \div 10 = \0.80 each. $\$0.75 < \0.80 , so 6 for \$4.50 is the better buy.

Answer: A. 6 for \$4.50 — \$0.75 each

Example 2

A faucet drips 15 ounces of water in 5 minutes. What is the unit rate?

Solution: $15 \div 5 = 3$ ounces per minute.

Answer: A. 3 ounces per minute

Example 3

A printer prints 42 pages in 6 minutes. How many pages does it print in 15 minutes at the same rate?

Solution: Unit rate: $42 \div 6 = 7$ pages per minute. In 15 minutes: $7 \times 15 = 105$ pages.

Answer: A. 105 pages

Write About the Math

The Writing Revolution

I can explain my comparison using the words unit rate, better buy, per unit, and rate.

1. Kernel Sentence subject + verb

Model: Unit Rate is a rate for just 1 of something, like cost for 1 item.

Tasa unitaria es una tasa para solo 1 de algo, como el precio de 1 artículo.

Write a kernel sentence about unit rate. Use a subject and a verb.

Escribe una oración base sobre tasa unitaria. Usa un sujeto y un verbo.

2. Sentence Expansion because • but • so

Kernel: Unit Rate matters in math

Tasa unitaria importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Unit Rate matters in math because ____.

Tasa unitaria importa en matemáticas porque ____.

but
pero

Unit Rate matters in math, but ____.

Tasa unitaria importa en matemáticas, pero ____.

so
entonces

Unit Rate matters in math, so ____.

Tasa unitaria importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement
Afirmación

Tell one true fact about unit rate.
Di un hecho verdadero sobre unit rate.

Unit rate ____.

Question
Pregunta

Ask a question about unit rate.
Haz una pregunta sobre unit rate.

How does ____ ?

¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about unit rate.
Muestra entusiasmo sobre unit rate.

Wow, ____ !

¡Guau, ____ !

Command
Mandato

Tell a partner what to do with unit rate.
Dile a un compañero qué hacer con unit rate.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

I can't compare the totals because ____.

No puedo comparar los totales porque ____.

Finding the cost ____ **lets me compare fairly.**

Hallar el costo ____ *me deja comparar de forma justa.*

The unit rate for ____ **is \$** ____ **per token.**

La tasa unitaria de ____ *es \$* ____ *por ficha.*

Try It

Solve on your own. Check the answer key when you are done.

1. A machine dispenses 24 tokens in 3 minutes. At this rate, how many tokens does it dispense in 10 minutes?

- A. 80 tokens
- B. 60 tokens
- C. 72 tokens
- D. 240 tokens

Show your work:

2. A factory produces 360 widgets in 8 hours. At the same rate, how many widgets are produced in a 12-hour shift?

- A. 540 widgets
- B. 480 widgets
- C. 360 widgets
- D. 720 widgets

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Two arcade supply companies offer token deals. Company X: 300 tokens for \$36. Company Y: 500 tokens for \$55. The arcade needs 1,000 tokens. Find the unit rate for each company, determine which is cheaper per token, and calculate the total cost of buying 1,000 tokens from the better deal.

Sentence starter: Company X charges \$___ per token and Company Y charges \$___ per token. The better deal is Company ___ because ___. Buying 1,000 tokens would cost \$___.

Show your work:

Reflect — Exit Ticket

A car travels 195 miles on 6 gallons of gas. What is the unit rate in miles per gallon?

- A. 32.5 miles per gallon
- B. 33 miles per gallon
- C. 30 miles per gallon
- D. 1,170 miles per gallon

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. 80 tokens — *Unit rate: $24 \div 3 = 8$ tokens per minute. In 10 minutes: $8 \times 10 = 80$ tokens.*
2. **Try It 2:** A. 540 widgets — *Unit rate: $360 \div 8 = 45$ widgets per hour. In 12 hours: $45 \times 12 = 540$ widgets.*
3. **Exit Ticket:** A. 32.5 miles per gallon — *$195 \div 6 = 32.5$ miles per gallon.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Unit Rate is a rate for just 1 of something, like cost for 1 item.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).